NLP 243 Deep Learning for Natural Language Processing

Fall 2024

Canvas: https://canvas.ucsc.edu/courses/76207

Introductions

Instructor: Dr. Amita Misra

Amazon: Lead Multi-modal and Search Amazon AGI: 2021-Present
IBM: Lead NLP Scientist NLU: 2018-221
UCSC: Adjunct professor: 2022-Present
amisra2@ucsc.edu
Office Hour: I will be available 30 minutes before the class or by appointment

TA: Sicong Huang
Section: Wednesday 1:30-3:05pm
Office Hours: Wednesday 3:15-4:15pm

You must attend the weekly sections and class. We will take attendance.

Goals of the class

- Gain experience on machine learning approaches for NLP and learn: How to frame NLP tasks with deep learning methods, focusing on sample classification and sequence classification
- Intro/reminder for general machine learning
- How deep learning works
- Using PyTorch to implement designed models
- When and why specific deep learning techniques work for specific problems

Expectations

- Your knowledge of Python is already pretty solid.
- You have basic knowledge of probability and linear algebra.
- We will go over examples written in PyTorch.
- There will be multiple coding assignments focusing on:
  - sequence classification (homeworks 1 and 2)
    - Movies directed by Cameron -> directed_by
  - sequence tagging (homework 3)
    - Movies directed by Cameron -> Movies directed by Cameron (director_name)
      Final project to demonstrate free form ideas.
- Final project to demonstrate free form ideas.

Outline for Today

Class Logistics

- What is natural language processing?
- What is machine learning?
- What is deep learning?

Class Web Page and Resources

Canvas:

- https://canvas.ucsc.edu/courses/
- Includes complete syllabus (but subject to change!)

Important Information

- Must attend at least 80% of classes. Let me know if you need to miss a class via email (amisra2@ucsc.edu)
- **Asking questions is encouraged!**
- We will have sections every week, highly recommend attending
- Three Homework Assignments
- One Team Project – SemEval 2025:
  - https://semeval.github.io/SemEval2025/tasks.html
- First 2-4 lectures will include a review of basic probability and linear algebra and pre-deep-learning machine learning methods.

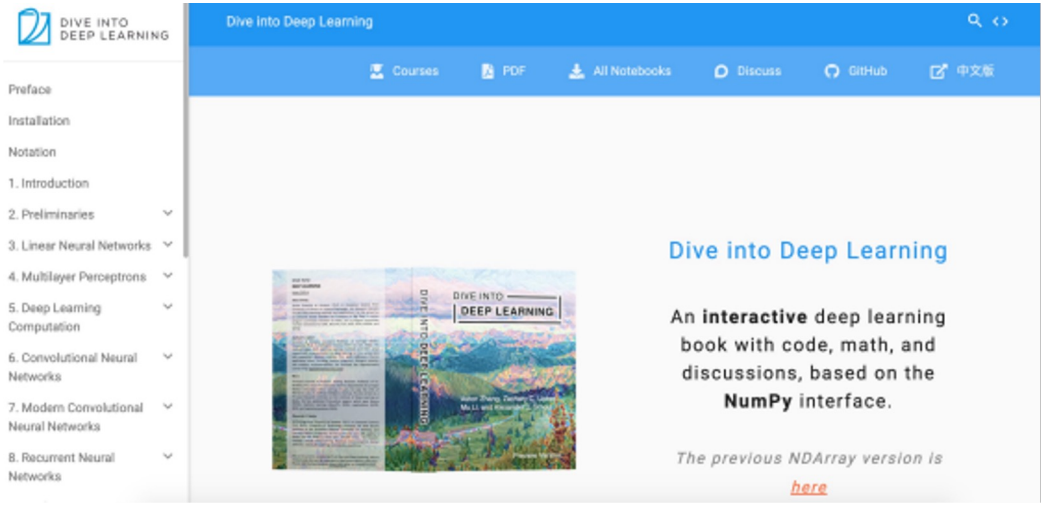
Important Information

- Special Accommodations:
  - *If you have special needs, we will accommodate you. The* **Disability Resource Center** *offers services that are confidential and free of charge.*
  - http://drc.ucsc.edu/
  - https://intranet.soe.ucsc.edu/DisabilityResource

1. Students contact the DRC to determine their eligibility
2. Students then notify their instructor and provide their Accommodation Authorization form.
3. Please note that it is the student's responsibility to contact the instructor about their accommodations.
4. Students should submit their requests to faculty no later than 7 days before a regular exam and 14 days before a final exam.

Books and other online resources

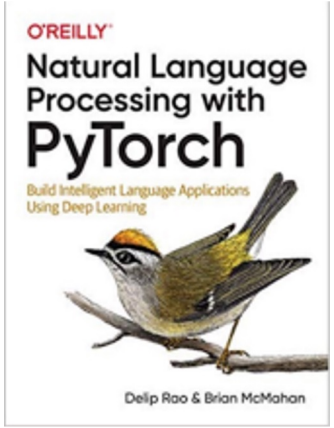
Dive Into Deep Learning, Ashton Zhang, Zack C. Lipton, Mu Li, Alex Smola. http://d2l.ai



Books and other online resources

Natural Language Processing with PyTorch. Delip Rao and Brian McMahan.

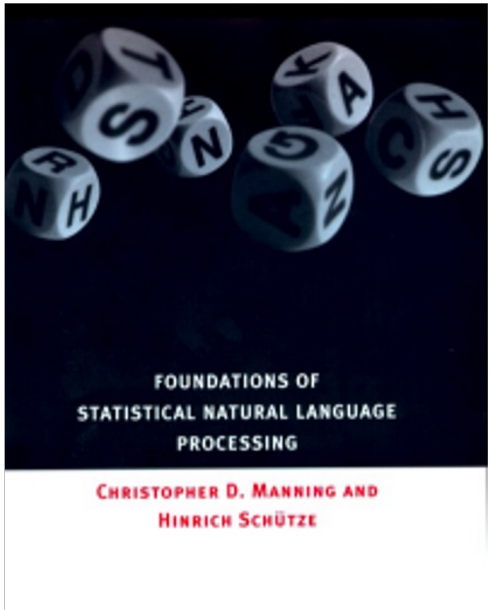
https://learning.oreilly.com/library/view/natural-language-processing/9781491978221/



Books and other online resources

Foundations of Statistical NLP. Chris Manning, Hinrich Schuetze.

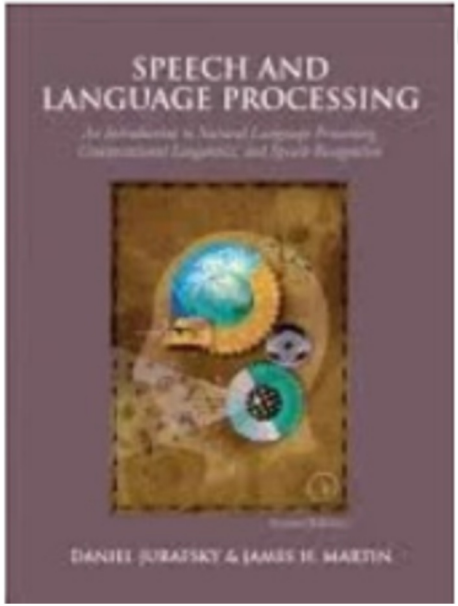
https://nlp.stanford.edu/fsnlp/



Books and other online resources

Speech and Language Processing, Dan Jurafsky and James Martin.

https://web.stanford.edu/~jurafsky/slp3/



Books and other online resources

Goodfellow, Bengio, and Courville, “Deep Learning,” 2016. http://www.deeplearningbook.org

- Michael Nielsen, “Neural Networks and Deep Learning”

http://neuralnetworksanddeeplearning.com

- Yoav Goldberg. A Primer on Neural Network Models for Natural Language Processing

- Please share other useful sites with the class on piazza!

Lecture Slides

In addition to my slides, I'll be using slides from:

- Dilek Hakkani-Tur, Amazon
- Yun-Nung (Vivian) Chen, NTU
- Marilyn Walker, UCSC
- Ian Lane, UCSC
- Kemal Oflazer, CMU
- Chris Manning, Stanford
- Mu Li and Alex Smola, AWS & UC Berkeley

Grading Policy

Attendance : 5%

- Homeworks and Project & Project presentations: 60%
- Midterm: 15%
- Final: 20%

- Homework Delivery:

We will organize one in-class competition and a leaderboard for each homework (i.e CodaLab) Every student should create a CodaLab account to participate in. In the CodaLab in-class competition, students are given the training data and labels. They need to train the requested models and submit their predictions on test data on CodaLab. CodaLab will rank their results according to evaluation metric (e.g. accuracy and F1 score).

Students also need to turn in one report (must be PDF only) and a zip file with their code on Canvas assignments.

Grades will consider both the ranking on leaderboard and the reports: 25% of grading will be based on performance on leaderboard, 50% will be based on the report accompanying the homework, and 25% will be based on the code.

Final Project

In Teams (no single person teams!)

- 10/10 In class, Possible final project ideas presentation
- 10/15 In class, final project team-up event
- 10/13 Register teams
- 10/31 Project proposals (1-2 page, 5 min presentation in class)
- 12/05 (Final, in class presentations, 10 min presentation + 2-3 min questions, 12-15 slides)
- 12/06 Final reports due (6-8 pages, conference paper style)

Outline for Today

- Class logistics
- **What is natural language processing?**
- What is machine learning?
- What is deep learning?

What is Natural Language Processing (NLP)?

- The field of natural language processing (NLP) develops **theories and computational models** of **natural language data** (the language that humans speak and write).

Why is NLP growing now?

- **So much NLP data**: online forum discussions, posts on Facebook, LinkedIn and other social sites, books, journal and newspaper articles, phone calls, radio broadcasts, YouTube user generated videos or public lectures, web pages, weblogs, tweets, user reviews, email, and comment streams on newspaper or magazine articles.
- **So many NLP applications**: Education, Economics, Digital Humanities, Biological, Environmental, Medical and Health Informatics, Conversational Interfaces, Search, Indexing, Business intelligence, Translation, Computational advertising.
- **So many interested parties:** Research, companies and government funding agencies, as well as an area of inherent theoretical interest.

Why Deep Learning for NLP?

- **So much NLP data**
- **Machine Learning approaches especially** deep learning methods **have dominated the field over the past decade**

Given the success of ML approaches, main NLP conferences such as ACL, NAACL, EMNLP, and others include mainly ML related papers.

Outline for Today

- Class logistics
- What is natural language processing?
- **What is machine learning?**
- What is deep learning?

What is Machine Learning?

- **Machine learning** (**ML**) is the field of artificial intelligence that develops **algorithms and statistical models** that **computers** use to **perform a specific task** without using explicit instructions (i.e., coding the specific steps for performing the task), relying on patterns and inference instead.

Example Task: Sentiment Detection for Reviews

- Program for solving tasks:

Input: "I love this product!"

↓ `program.py`

+

if input contains "love", "like", etc.
output = positive

"It claims too much."

↓ `program.py`

-

if input contains "too much", "bad", etc.
output = negative

"It's a little expensive."

↓ `program.py`

?

Some tasks are complex, and we don't know how to write a program to solve them
Example: speech recognition.

Example Task: Sentiment Detection for Reviews

- Learning ≈ Looking for a Function

Input: "I love this product!" ↓ f +

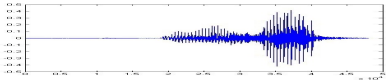
"It claims too much." ↓ f -

"It's a little expensive." ↓ f ?

Given a large amount of data, the machine learns what the function f should be.

Other Example Tasks

- Speech Recognition

f  “the book”

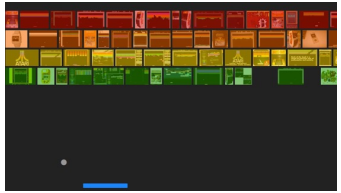
- Handwriting Recognition

f “2”

- Weather forecast

f Thursday “ Saturday”

- Playing video games

f  “move left”

Other Examples: NLP Tasks

- Machine Translation

f “Kitap masada.” “The book is on the table.”

- Question Answering

f “Who wrote Leviathan?” “Paul Auster”

- Summarization

f

Kinds of Machine Learning

- Supervised learning
- Unsupervised learning
- Reinforcement learning

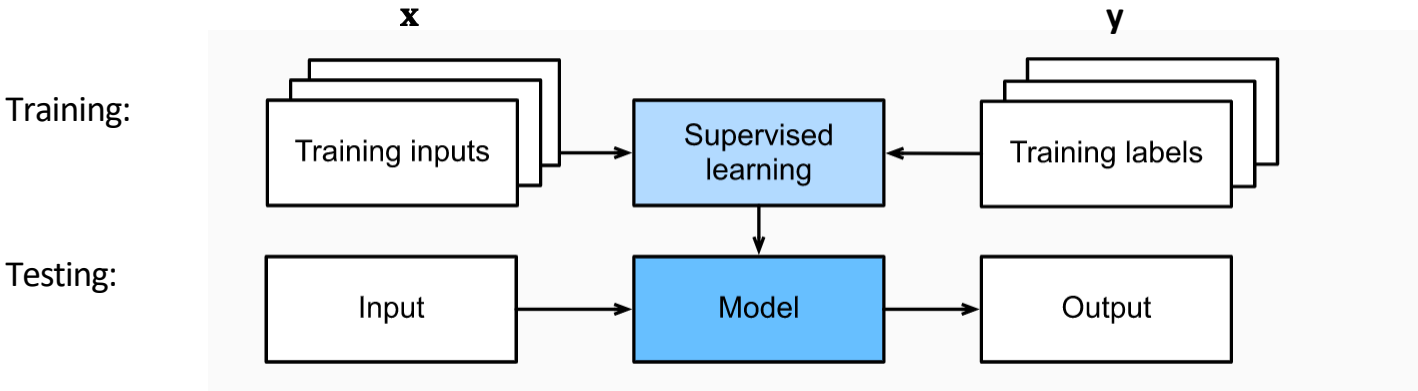
Supervised Learning

- Finding functions for predicting **targets** given inputs from pairs of **inputs and targets**
- The targets, which we often call ***labels***, are generally denoted by $\boldsymbol{y}$
- The input data, also called the *features* or covariates, are typically denoted $\mathbf{x}$

Supervised Learning

x : "It claims too much."
function input

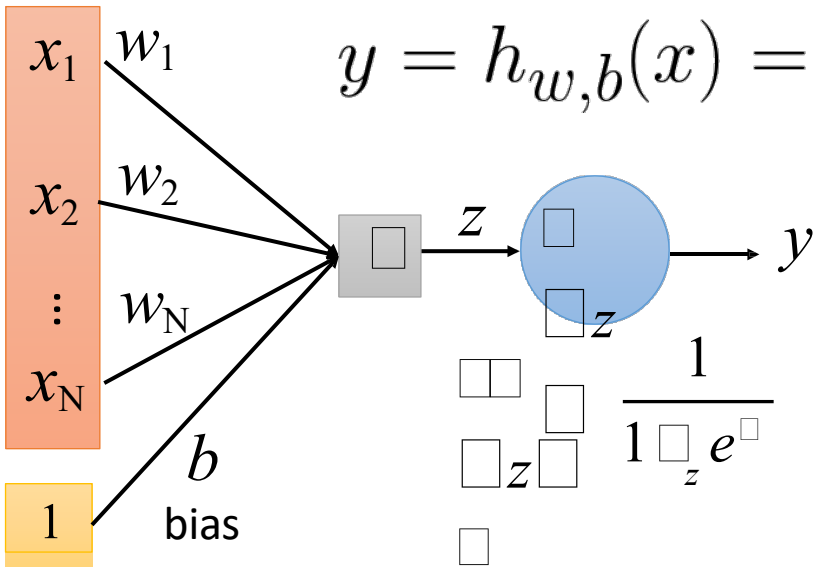
y: - (negative)
function output



Supervised Learning

- Each (input, target) pair is called a ***example*** or an ***instance***
- Supervised learning learns the model/function from examples
- For example, machine translation models are trained from pairs of source and target language sentences

Example: Model Parameters of A Single Neuron



$$y = h_{w,b}(x) = \sigma(w^T x + b)$$

w, b are the parameters of this neuron

Kinds of Machine Learning

- Supervised learning
- Unsupervised learning
- Reinforcement learning

Unsupervised Learning

- Finding previously unknown patterns in data set without pre-existing labels.
  - Clustering
  - Zero-Shot Learning
  - Few-Shot Learning
  - Semi-supervised Learning
  - Weakly-supervised Learning
  - Transfer Learning

Kinds of Machine Learning

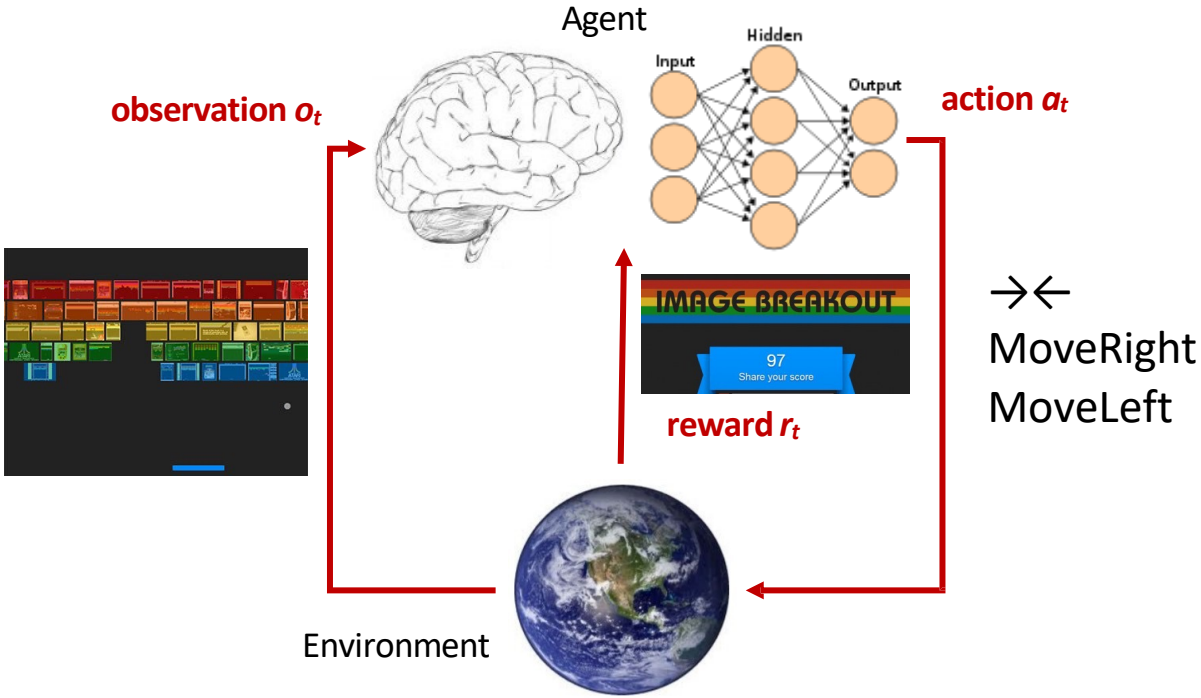
- Unsupervised learning
- Supervised learning
- Reinforcement learning

Reinforcement Learning

- Learning action prediction policies through interactions with an environment over a series of *timesteps*.
  - At each timestep t, the learning agent receives some observation o_t from the environment and must choose an action a_t that is subsequently transmitted back to the environment via some mechanism.
  - Finally, the agent receives a reward r_t from the environment.
  - The agent then receives a subsequent observation, and chooses a subsequent action, and so on.
  - A *policy* is a function that maps from observations to actions.

Agent and Environment

- No labels, just a reward
- Agent can exploit and explore
- More on RL in NLP 244!



→←
MoveRight
MoveLeft

Kinds of Machine Learning

- Supervised learning
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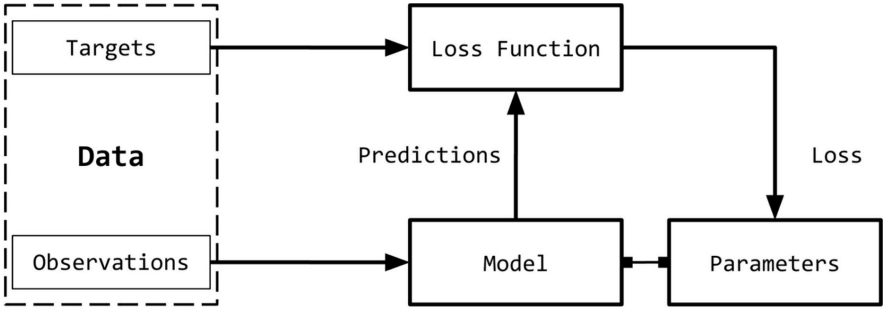
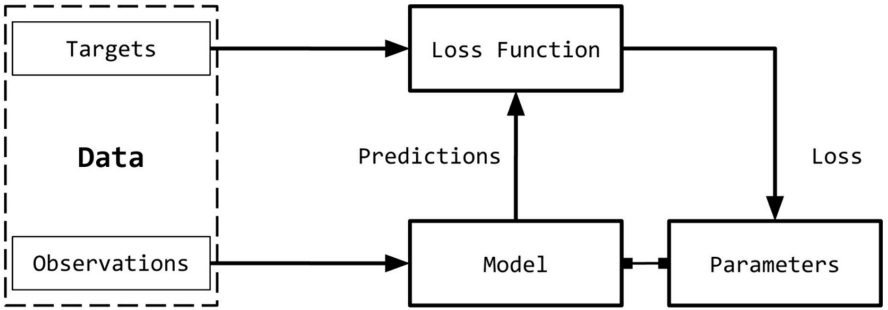
Terminology

- **Observations**: inputs, x.
- **Targets**: labels, y, corresponding to the observations (also called ground truth sometimes).
- **Model**: Function that takes an observation and predicts the value of its target label.



More Terminology

- **Parameters**: weights (for deep learning), w, that parameterize the model.
- **Predictions**: Also called estimates, target values estimated by the model, $\hat{y}$.
- **Loss Function**: a function (denoted L) that compares how far off a prediction is from its target. The lower the loss, the better is the model at predicting the target.



Supervised Learning - Regression

- When the targets take arbitrary values in a range (instead of specific categories such as +/- for sentiment classification), we call it a **regression** problem.
- Examples:
  - Estimating the price of a house given attributes such as location, square footage, number of rooms, etc.
  - Predicting how many miles can a dog run given attributes such as its breed, age, size, etc.

Supervised Learning - Classification

- When the targets take a limited number of pre-defined categories, we call it a classification problem.
- **Binary classification:** only two values
  - Positive/negative sentiment classification
  - Sentence segmentation from speech: each word boundary is classified into sentence boundary versus not a sentence boundary.
  - Sentence tokenization for text: each "." is classified into two classes similarly.
- **Multi-class classification:** more than two values
  - Digit recognition: each digit is classified into one of 10 values for 0,…,9.

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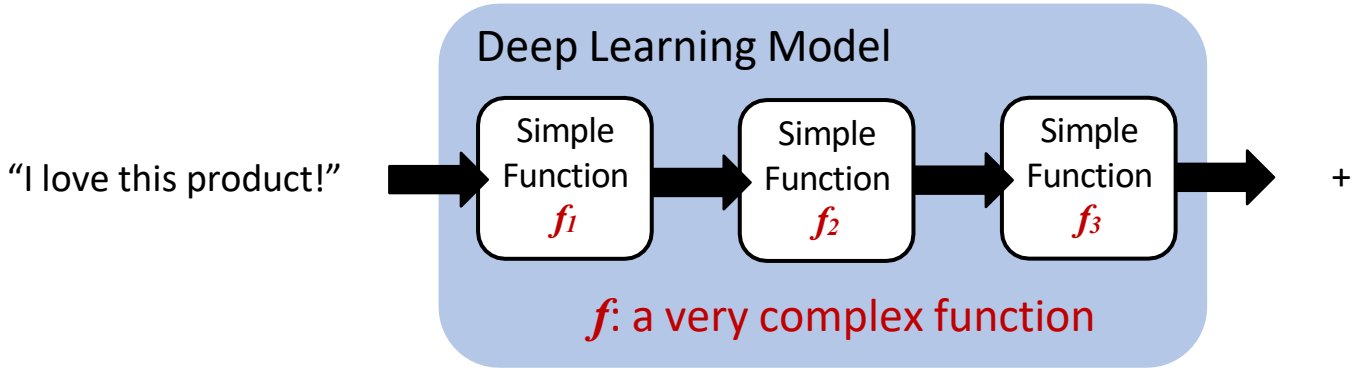
- What is machine learning?
- **What is deep learning?**

Stacked Functions Learned by Machine

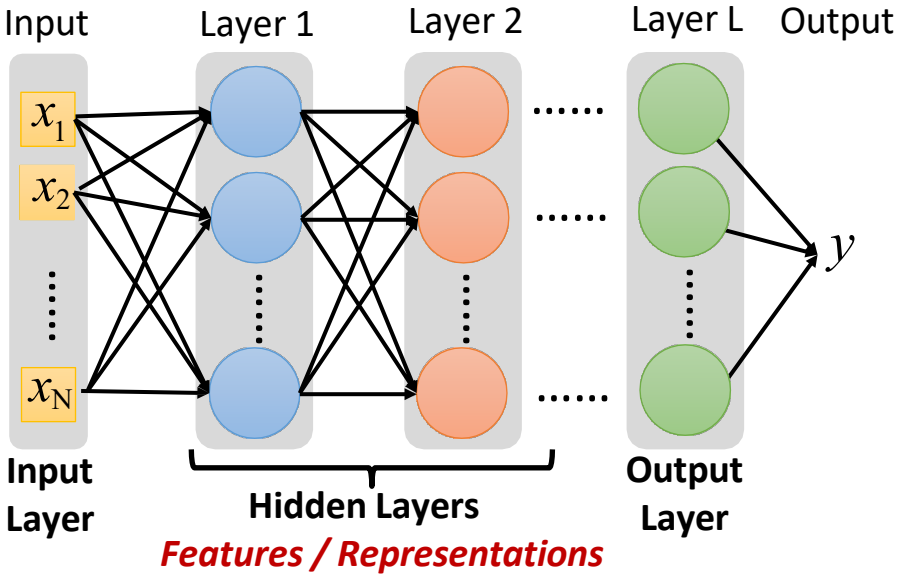


End-to-end training: what each function should do is learned automatically

Deep learning usually refers to *neural network* based model

Stacked Functions Learned by Machine

vector $\boldsymbol{x}$
"I love this product."



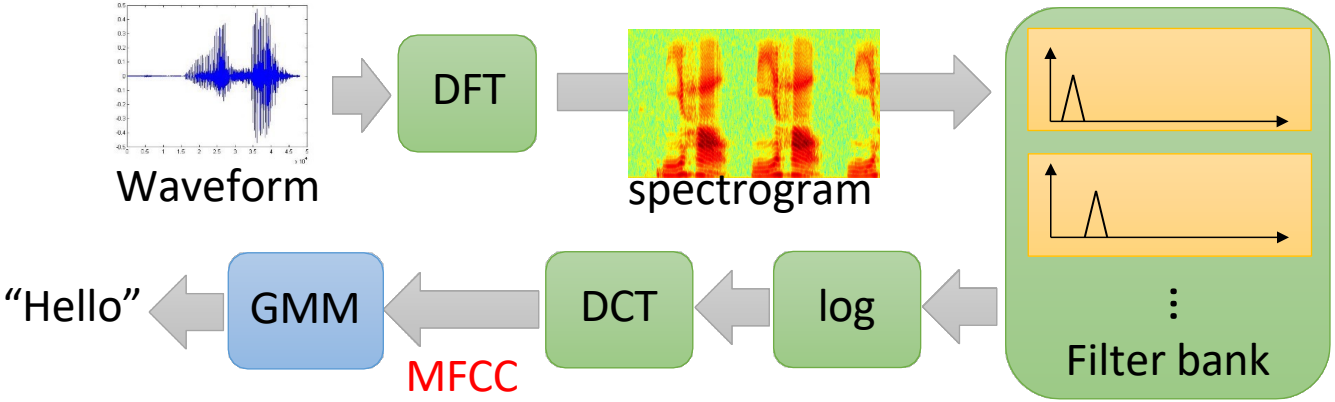
label $\boldsymbol{y}$
+

Representation Learning attempts to learn good features/representations

Deep Learning attempts to learn (multiple levels of) representations and an output

Deep vs. Shallow – Speech Recognition

- Shallow Model

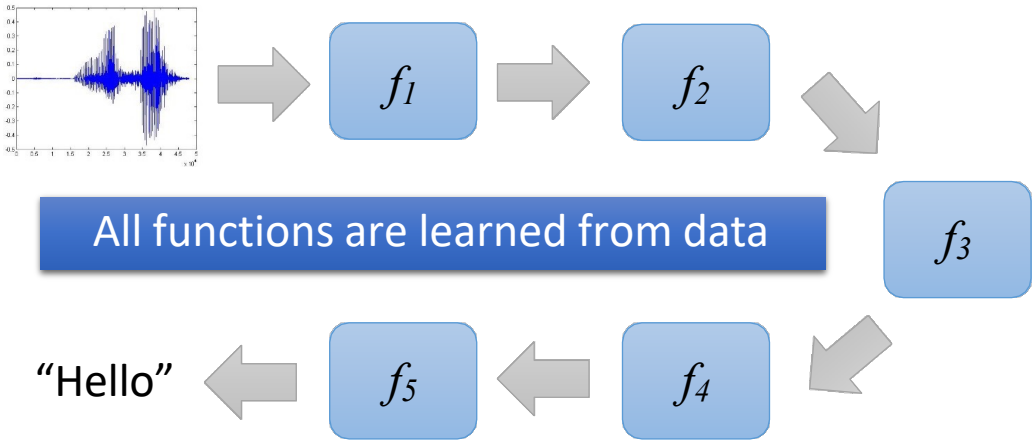


Each box is a simple function in the production line:

:hand-crafted :learned from data

Deep vs. Shallow – Speech Recognition

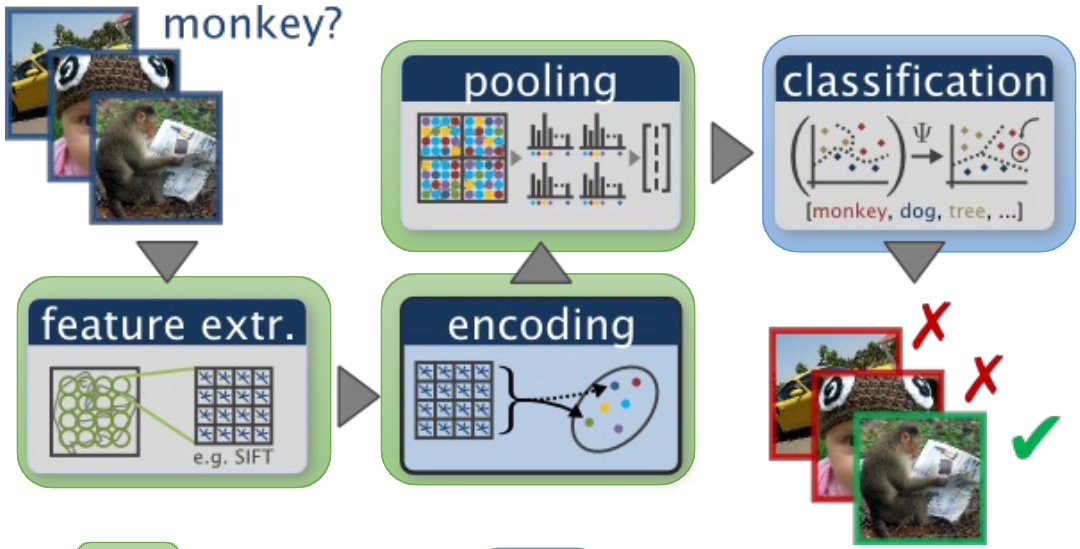
- Deep Model



Less engineering labor, but machine learns more

Deep vs. Shallow – Image Recognition

- Shallow Model



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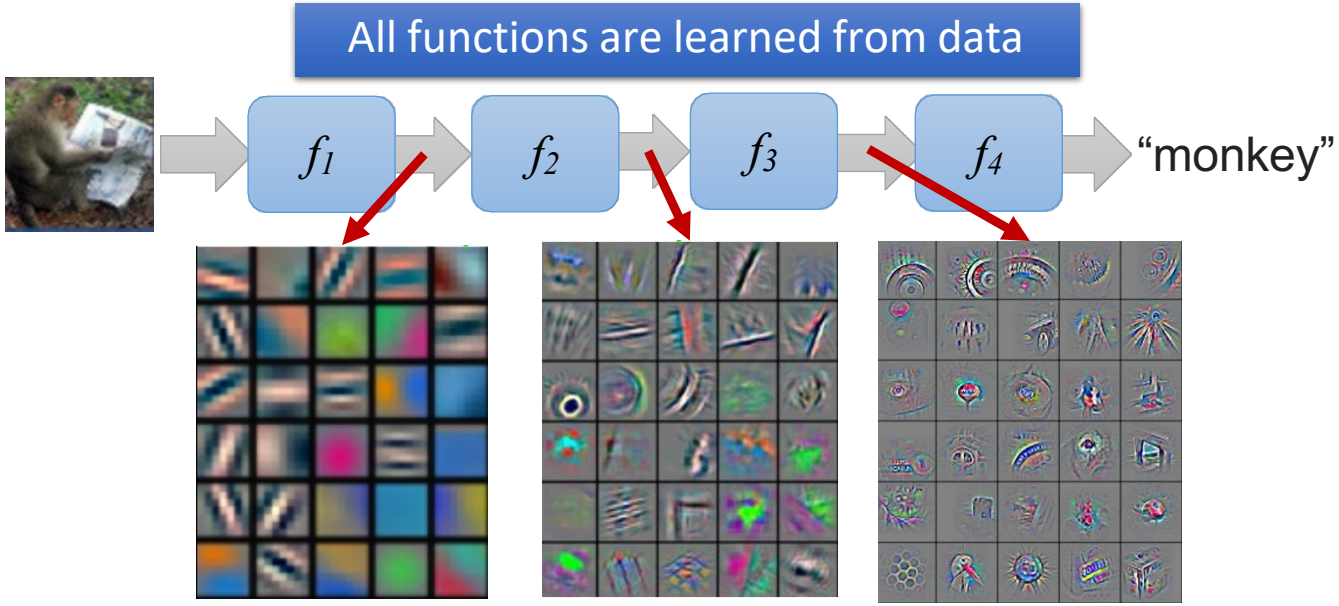
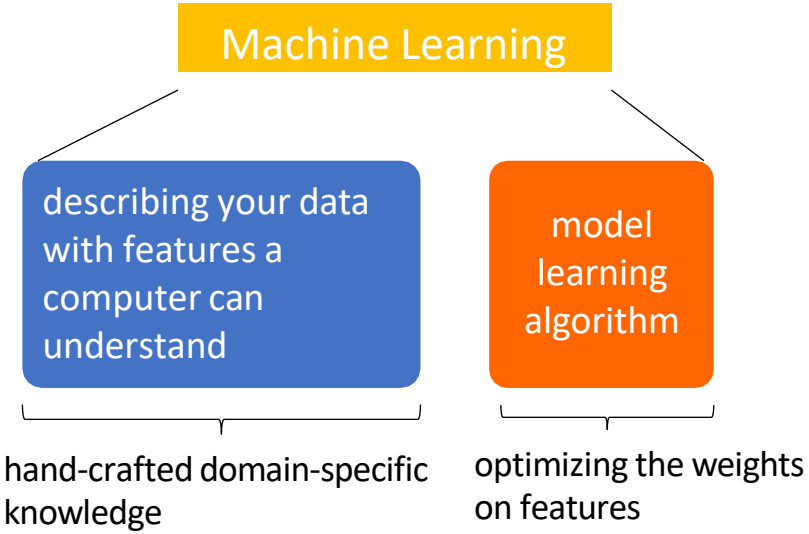
Deep vs. Shallow – Image Recognition

- Deep Model

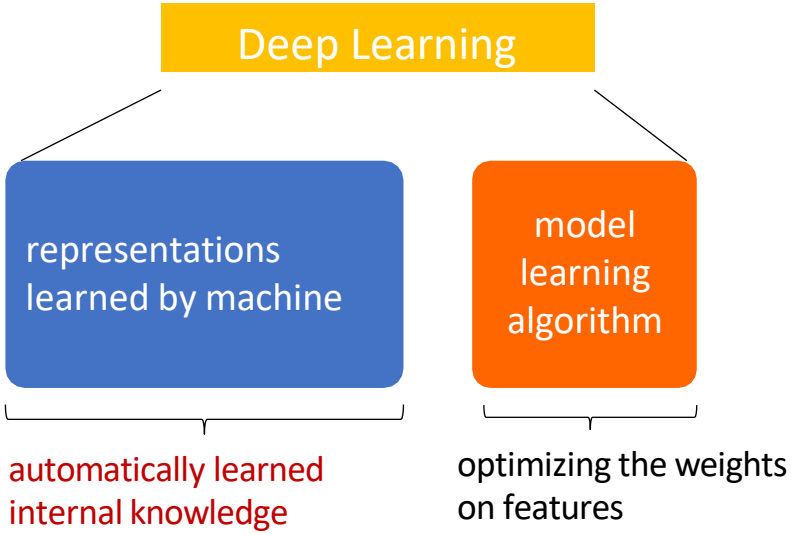


Features / Representations

Machine Learning vs. Deep Learning

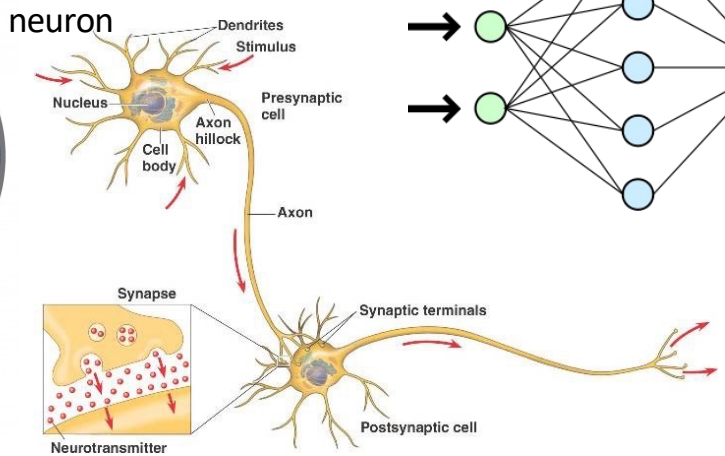


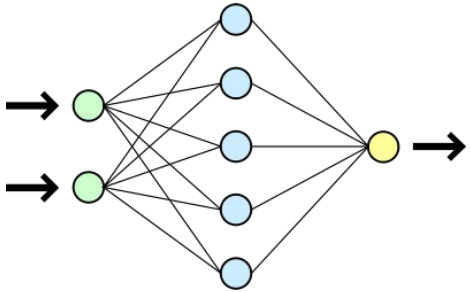
Machine Learning vs. Deep Learning



Deep learning usually refers to *neural network* based model

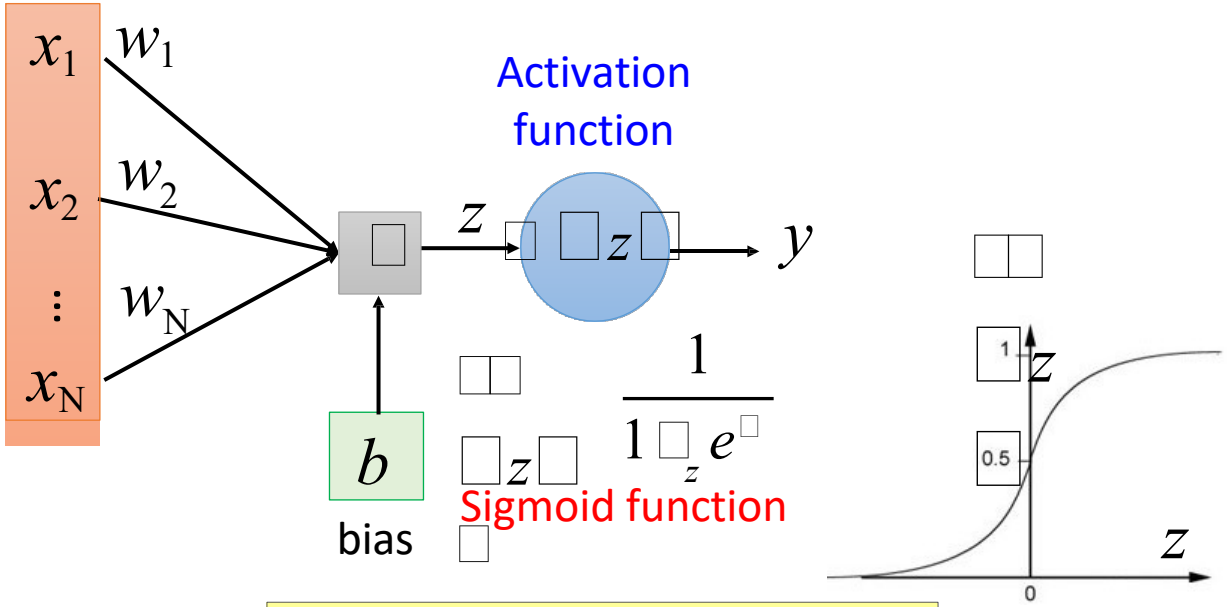
Inspired by Human Brain





Neurotransmission section of: https://en.wikipedia.org/wiki/Human_brain

A Single Neuron



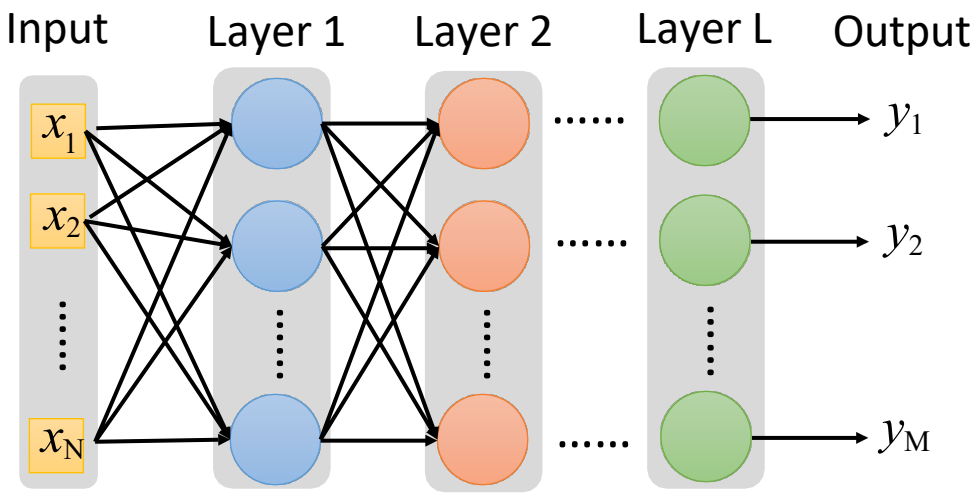
Each neuron is a very simple function

Deep Neural Network

A neural network is a complex function:

$$f : R^N \square R^M$$

- Cascading the neurons to form a neural network



Each layer is a simple function in the production line

Suggestions and Readings

- Attend the NLP Bootcamp tomorrow morning 10:30am RM2227
- Polish your Jupyter/Colab notebook skills
- Get started with PyTorch

- Readings:
  - Ch1 of NLP with PyTorch
  - Ch1 of Dive into Deep Learning

PyTorch

- Open-source, community driven deep learning framework.
- https://pytorch.org/
- Information on getting started locally.
- Also available on colab.

```
import torch
import numpy as np
```

Questions?